

Madhav Dhakal

Assistant Research Professor

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Career Objective

To secure a challenging position that utilizes my 10+ years of experience in agricultural research, teaching, and development, while allowing me the opportunity to enhance my specialized skills in a dynamic, inclusive, and stable workplace.

Education

Ph.D. in Plant and Soil with emphasis in Forage Agronomy and Soil Water Dynamics
Texas Tech University, Lubbock, TX. Advisor: Dr. Chuck West. May 2019. GPA 3.957/4.00

Master of Science in Agriculture with emphasis in Weed Science
Tribhuvan University, Nepal. Advisor: Dr. S.K. Sah. August 2011. Distinction (86.44%)

Bachelor of Science in General Agriculture
Tribhuvan University, Nepal. August 2009. Distinction (81.89%)

Core Competencies and Technical Skills

- Expert in Microsoft Word, Excel, PowerPoint, Adobe Creative Cloud applications
- Expert in data science: Python (PyCharm & VSC), R-Studio, SAS, MATLAB, and LaTeX
- Proficient in geospatial tools: ArcGIS, ERDAS Imagine, ENVI, QGIS, Google Earth Engine
- Expert in field instrumentations: Eddy Covariance, irrigation system, soil moisture sensors
- Handle 13 analytical equipment including Liquid/Gas Chromatography and CN analyzer
- Physical and hydraulic models: HYDRUS, RZWQM2, AGNPS, and DSSAT
- Design statistically fit and objective-driven biological and environmental research
- Teach undergraduate and graduate courses in agronomy and soil sciences
- Secure federal and state funding, handle multiple research projects, and lead large teams
- Handle and analyze complex datasets, publish research papers, write project reports

Professional Experiences

Assistant Research Professor

Mississippi State University, Starkville, MS. December 2023 – Present

- Lead soil water-plant-atmospheric research, including agrohydrology and agroecology
- Identify the key factors affecting biogeochemical processes in agricultural systems
- Mentor graduate and undergraduate students, publish data, and secure & manage grants
- Cultivate and manage strategic relationships with MSU's diverse stakeholders

Research Scientist

Rodale Institute, Kutztown, PA. January 2022 – November 2023

- Developed row crop and weed management program in regenerative organic systems
- Oversaw funded projects, managed external grants, and secured nearly 0.5M dollar grants
- Published peer-reviewed research papers, book chapter, and web articles on organic crops
- Led outreach activities that include field days, webinars, farm tours, and workshops

Postdoctoral Associate

USDA-Agricultural Research Service, Stoneville, MS. June 2019 – December 2021

- Managed Long-Term Agroecosystem Research (LTAR) in cotton and sorghum production
- Published high-impact peer-reviewed journal articles on soil hydrology and remote sensing

Graduate Research Assistant

Texas Tech University, Lubbock, TX. August 2015 – May 2019

- Determined water footprint for crude protein and digestible organic matter production in alfalfa-native grass mixtures for the Southern High Plains region of Texas
- Developed population decay models for alfalfa & calibration parameters for PR2/6 sensor

- Published high-impact research papers (two outstanding awards) in peer-reviewed journals

Horticulture Development Officer

Ministry of Agriculture Development, Udayapur and Kathmandu, Nepal, 2011 to 2015

- Designed and executed farmers' education programs including training, publication of extension materials, translational research and demonstration, and on-call services
- Supervised field extension agents and led capacity building trainings

Lecturer

Tribhuvan University, Institute of Agriculture and Animal Science, Baitadi, Nepal, 2010-2011

- Taught undergraduate agricultural courses including Principles of Agronomy AGR 111 (3+1), Seed Technology AGR 212 (3+1), and Introductory Soil Science SOL 211 (3+1)

Awards and Honors

- i. **'2025 Conservation Research Award'**, from Soil and Water Conservation Society.
- ii. **'2022 Agronomy Journal (AJ) Outstanding Associate Editor'**, from American Society of Agronomy.
- iii. **2021 ASA, CSSA, and SSSA Photo Contest** – "The Creation" won second place in the SSSA category.
- iv. **'Crop Science Outstanding Paper Award 2020'** – C6 Division: link to the paper: doi:10.1002/csc2.20159, by Crop Science Society of America.
- v. **'2020 Agronomy Journal (AJ) Outstanding Reviewer'**, from American Society of Agronomy.
- vi. **'Crop Science Outstanding Paper Award 2019'**, for a paper published in 2019: doi: 10.2135/cropsci2019.03.0156, by Crop Science Society of America.
- vii. **George Tereshkovich Plant and Soil Science Outstanding Ph.D. Graduate Award 2019/2020**, awarded by the Department of Plant and Soil Science, Texas Tech University, USA.
- viii. Second place in the **"Three Minute Thesis"** competition organized by the Graduate School, Texas Tech University on 24 October 2018.
- ix. **Graduate School Water Conservation Research Scholarship 2018/2019** from the Sandy Land Underground Water Conservation District, TX, USA for AY 2018 – 2019.
- x. **Recognition of Outstanding Communication**, Center for Agri-Science Communications, Texas Tech University, USA.
- xi. **James A. "Buddy" Davidson Foundation Water Conservation Fellowship Endowment (2018 – 2019)** by the Department of Plant and Soil Science, Texas Tech University, USA.
- xii. **'2016 Future Leader Fellow'** awarded by the Association of International Agriculture and Rural Development, Washington DC.
- xiii. **Harold & Mary Dregne Graduate Program Endowment Fellowship** for the 2015 – 2018 Academic school years provided by the Department of Plant and Soil Science.
- xiv. **Graduate Research Assistantship** for Ph.D. research from August 2015 to May 2019.
- xv. **Research Fellowship (2010 – 2011)** by International Maize and Wheat Improvement Center (CIMMYT), Kathmandu, Nepal for MS thesis research.
- xvi. **Merit Scholarship** from Tribhuvan University, Nepal for the study of MS and BS in 2009 and 2005, respectively.
- xvii. **'Gold Medal'** and a full-tuition scholarship from Saptagandaki Multiple Campus, Bharatpur, Chitwan, Nepal.

Publications

A. Peer-reviewed journal articles published or accepted:

1. **Dhakal, M.**, D. Panday, F. Etemadi, R.K. Afshar, C. Lapham, A. Smith, A. Ghale. (2026). Tillage and cover cropping enhance yield, nitrogen use efficiency, and soil health in organic industrial hemp-barley rotations. *Agron. J.* 118:e70306. <https://doi.org/10.1002/agj2.70306>
2. Panday, D., P. Heydar, C. Lapham, A. Pierce, **M. Dhakal**, A. Ghale, G. Fioravanti, R. Kander. (2026). Mechanical quality of hemp fiber as influenced by tillage, cover crop, and nitrogen management in regenerative organic systems. *Ind. Crops Products* 239(2026). <https://doi.org/10.1016/j.indcrop.2025.122466>
3. **Dhakal, M.**, M.A. Locke, K.N. Reddy, M.T. Moore, R W. Steinriede Jr., T.P. Kharel, L J. Krutz. (2025). Soil-plant-water relations and water footprint of cover crop-based no-till cotton and sorghum productions in humid region. *Soil Sci. Soc. Ame. J.* DOI: 10.1002/saj2.70166
4. Panday, D., B. Acharya, **M. Dhakal**, T. Caton, C. Lapham, A. Smith, A. Ghalegholabbehbahani. (2025). Industrial hemp yield and chemical composition as influenced by row spacing, fertilization and environmental conditions. *Agrosyst. Geosci. Environ.* 8:e70093. DOI: 10.1002/agg2.7009
5. Ghalegholabbehbahani, A., G. Zinati, S. Hamido, N. Bhusal, **M. Dhakal**, R. Afshar, J.B. Contina, R. Caetani, A.H. Smith, D. Panday. (2025) Integrated Control of Powdery Mildew Using UV Light Exposure and OMRI Certified Fungicide for Greenhouse Organic Lettuce Production. *Horticulturae* 11, 246. <https://doi.org/10.3390/horticulturae11030246>
6. Kharel, T., H. Tyler, P. Mubvumba, Y. Huang, A. Bhandari, R. Fletcher, S. Anapalli, D. Joshi, A. Mengistu, G. Birru, K. Adhikari, **M. Dhakal**, M. Maskey, K. Reddy, D. Clay. (2025). Machine learning on multi-spectral imagery to estimate nutrient yield of mixed-species cover crops. *Agric. Environ. Letters* 10:e70009. <https://doi.org/10.1002/ael2.70009>
7. **Dhakal, M.**, M.A. Locke, K.N. Reddy, M.T. Moore, R W. Steinriede, Jr. L J. Krutz. (2024). Improving soil water storage with no-tillage cover cropping in the Mississippi River Alluvial Basin. *Soil Sci. Soc. Ame. J.* 88:540-556. 10.1002/saj2.20638
8. **Dhakal, M.**, Y. Rui, A.R. Benson, P.O. Hinson, K. Delete, R.K. Afshar, B. Luck, A. Smith. 2024. Cover crop management strategies affect weeds and profitability of organic no-till soybean. *Renew. Agric. Food Syst.* 39:e3. <https://doi.org/10.1017/S1742170523000522>
9. Kharel, G., **M. Dhakal**, S.K. Deb, L. Slaughter, C. Simpson, and C.P. West. 2023. Effect of long-term pasture management on soil hydraulic and thermal properties. *Plants* 12, 1491. <https://doi.org/10.3390/plants12071491>
10. **Dhakal, M.**, Y. Huang, M.A. Locke, K. Reddy, M. Moore, J. Krutz, G. Drew, and R. Bajgaighn. 2022. Assessment of cotton and sorghum stand establishment using UAV-based multispectral and DSLR-based RGB imagery. *Agrosys. Geosci. Environ.*, 5, e20247. doi:10.1002/agg2.20247
11. Singh, G., **M. Dhakal**, A. Sharma, J.E. Schoonover, K.W.J. Williard, and G. Kaur. 2022. Cover Crops and Landscape Positions mediate Corn-Soybean Production. *Agrosys. Geosci. Environ.*, 5, e20249. doi:10.1002/agg2.20249
12. Singh, G., **M. Dhakal**, L. Yang, G. Kaur, K.W. Williard, J.E. Schoonover, and A. Sadeghpour. 2020. Decomposition and nitrogen release of cereal rye and hairy vetch in conventional and no-tillage systems. *Agron. J.* 112:3605–3618. doi:10.1002/agj2.20268
13. **Dhakal, M.**, C.P. West, C. Villalobos, J.O. Sarturi, and S.K. Deb. 2020. Trade-off between nutritive value improvement and crop water use for alfalfa-grass system. *Crop Sci.* 60:1711-1723. doi:10.1002/csc2.20159

14. **Dhakal, M.**, G. Singh, R.L. Cook, and T. Sievers. 2020. Modeling hairy vetch and cereal rye cover crop decomposition and nitrogen release. *Agronomy*. 10(5), 701. doi.org/10.3390/agronomy10050701
15. **Dhakal, M.**, C.P. West, C. Villalobos, P. Brown, P.E. Green. 2020. Interseeding alfalfa into native grassland for enhanced yield and water use efficiency. *Agron. J.* 112:1931-1942. doi:10.1002/agj2.20147
16. **Dhakal M.**, C.P. West, S.K. Deb, C. Villalobos, and G. Kharel. 2020. Row spacing of alfalfa interseeded into native grass pasture influences soil-plant-water relations. *Agron. J.* 112:274-287. doi:10.1002/agj2.20012
17. **Dhakal, M.**, C.P. West, and C. Villalobos. 2019. Establishment and stand development of alfalfa interseeded into native grass mixture: Cultivar and row spacing effects. *Crop Sci.* 59:2271–2279. doi:10.2135/cropsci2019.03.0156
18. **Dhakal, M.**, C.P. West, S.K. Deb, G. Kharel, and G.L. Ritchie. 2019. Field calibration of PR2 capacitance probe in Pullman clay-loam soil of Southern High Plains. *Agrosyst. Geosci. Environ.* 2:180043 doi:10.2134/age2018.10.0043
19. **Dhakal, M.**, S.K. Sah, and G. Kharel. 2019. Integrated weed management in direct-seeded rice: Dynamics and economics. *Int. J. Agric. Food Sci.* 3:83–92. doi:10.31015/jaefs.2019.2.6
20. **Dhakal, M.**, S.K. Shah, A. McDonald, and A.P. Regmi. 2015. Perception and economics of dry direct-seeded rice in Terai of Nepal. *J. Agric. Environ.* 16:103–111. doi:10.3126/aej.v16i0.19843

B. Book chapter published

1. **Dhakal, M.**, G. Zinati, M. Fulcher, D. Fornara, E. Martani, J.B. Contina, P. Hinson, R. Afshar, R. Ghimire. 2022. Challenges and emerging opportunities for weed management in organic agriculture. *In: D. Sparks (editor), Advances in Agronomy*, 184:125-172. <https://doi.org/10.1016/bs.agron.2023.11.002>

C. Web Articles published

1. Pandey, D., B. S. Acharya, **M. Dhakal**, T. Caton, C. Lapham, A. Smith, A. Ghalehgolabbabani. 2025. Industrial hemp yield and chemical composition as influenced by row spacing, fertilization, and environmental conditions. Research Article, Rodale Institute, Kutztown, PA. <https://rodaleinstitute.org/science/articles/>
2. Pandey, D., B. S. Acharya, **M. Dhakal**, T. Caton, C. Lapham, A. Smith, A. Ghalehgolabbabani. 2025. Improving industrial hemp yield and quality, what we learned about spacing, fertilizer, and environmental conditions. Research Article, Rodale Institute, Kutztown, PA. <https://rodaleinstitute.org/science/industrial-hemp-trial/>
3. Ghalehgolabbabani, A., J.B. Contina, **M. Dhakal**, A. Smith, D. Panday. 2024. Integrated disease management strategy to enhance microgreen production in regenerative organic systems. Research Article, Rodale Institute, Kutztown, PA. <https://rodaleinstitute.org/science/articles/>
4. **Dhakal, M.**, C. Lapham, A. Ghale, A. Smith. 2023. Growing industrial hemp as an organic cash crop. Web Article, Rodale Institute, Kutztown, PA. <https://rodaleinstitute.org/science/articles/growing-industrial-hemp-as-an-organic-cash-crop/>
5. **Dhakal, M.**, Q.V. Burren, C. Lapham. 2022. Nutrient management in cover crop-based organic hemp production. Newsletter, Rodale Institute, Pocono Organic Center, Long Pond,

PA. https://www.poconoorganics.com/s/Industrial_Hemp_Study_Details_August16.pdf

6. **Dhakai, M.**, C. West, C. Villalobos, P. Brown, P. Green, R. Kellison, K. Radicke, J.O. Sarturi, and D. McCallister. 2020. Improving grassland with alfalfa in the Texas High Plains. Texas Alliance for Water Conservation.
http://www.depts.ttu.edu/tawc/researchuploads/alfalfa_2020.pdf

C. Manuscripts submitted

1. Fleming, D.E., **M. Dhakai**, G.D. Spencer. (2025). Cover crop effects in Mississippi portal soybean production, part 2: soil organic carbon. J. Env't. Qual.
2. Fleming, D.E., **M. Dhakai**, L.J. Krutz, G.D. Spencer. (2026). Cover crops effects on soybean yield in the Mississippi portal of the USA. Agron. J.
3. Shrestha, A., T.P. Kharel, **M. Dhakai**, K.N. Reddy. (2026). Assessing trends and spatial patterns in soybean phenology using satellite remote sensing in the Lower Mississippi River Basin. ISPRS J. Photo. Rem. Sens.

D. Abstracts Presented – Oral

1. **Dhakai, M.**, M.A. Locke, K. N. Reddy, M.T. Moore, Jr. R. W. Steinriede, L. J. Krutz. (2025). Building resilient agroecosystem from the ground up: Soil and water benefits of cover cropping and conservation tillage in a humid region. Annual CANVAS meeting, ASA-CSSA-SSSA. Nov 9-13, 2025, Salt Lake City, UT.
2. Panday, D., P. Heydar, C. Lapham, A. Pierce, **M. Dhakai**, A. Ghale, G. Fioravanti, R. Kander. (2025). Improving hemp fiber strength and flexibility with regenerative practices and minimal nitrogen. Annual CANVAS meeting, ASA-CSSA-SSSA. Nov 9-13, 2025, Salt Lake City, UT.
3. Shrestha, A., T.P. Kharel, **M. Dhakai**, K.N. Reddy. 2025. Detecting soybean emergence and senescence with satellite remote sensing in the Lower Mississippi River Basin. CANVAS annual meeting, ASA-CSSA-SSSA. Nov 9-13, 2025, Salt Lake City, UT.
4. Oloyede, O., P.P. Joshi, G.D. Spencer, L.J. Krutz, P. Wagle, **M. Dhakai**. (2025). Effect of tailwater recirculation on water and methane fluxes in irrigated rice production system. Annual CANVAS meeting, ASA-CSSA-SSSA meeting, Salt Lake City, Nov 9-12, 2025 (*Second place*).
5. Joshi, P.P., Oloyede, O., Spencer, D., & **Dhakai, M.** (2025). Effect of Regenerative Management Practices on Greenhouse Gases in Corn. Annual CANVAS meeting, ASA-CSSA-SSSA meeting, Salt Lake City, Nov 9-12, 2025.
6. **Dhakai, M.**, M.A. Locke, K. N. Reddy, M.T. Moore, Jr. R. W. Steinriede, L. J. Krutz. (2025). Integration of regenerative management practices for improved soil health, water quality, and water use efficiency in cotton and sorghum cropping systems. Mississippi Water & Energy Conference. Oct 28-30, 2025, Starkville, MS.
7. Oloyede, O. E., Spencer, D., Joshi, P. P., Krutz, L. J., & **Dhakai, M.** (2025, October 30). Effect of tailwater recirculation on water and methane fluxes in irrigated rice production systems. Mississippi Water and Energy Conference, Mississippi Water Resources Research Institute, Mississippi State University, Mississippi.
8. Oloyede, O. E., Spencer, D., Joshi, P. P., Krutz, L. J., & **Dhakai, M.** (2025, October 18). Can tailwater recirculation in row rice reduce water and methane fluxes without a yield penalty? Graduate Research Fall Symposium, Graduate Student Association, Mississippi State University, Mississippi (*Second Place winner*).

9. Joshi, P.P., Oloyede, O., Spencer, D., & **Dhakar, M.**, (2025, October 18). Mitigating Greenhouse Gas Emissions from Corn Through Protected Fertility Management. Fall Graduate Symposium, Graduate Student Association, Mississippi State University, Mississippi
10. Joshi, P.P., & **Dhakar, M.** (2025, March 20). Improved fertility management reduces greenhouse gas emissions in corn. Mississippi Academy of Sciences (MAS), Biloxi, Mississippi.
11. **Dhakar, M.**, M.A. Locke, K. N. Reddy, M.T. Moore, Jr. R. W. Steinriede, L. J. Krutz. (2025). Impacts of cover cropping and tillage on soil health and root-zone water quality in cotton and sorghum production systems. Soil Water Conservation Society Annual meeting, Aug 6-8, 2025. Costa Mesa, CA.
12. **Dhakar, M.**, M.A. Locke, K.N. Reddy, M. Moore, W. Steinriede, J. Krutz. 2024. Soil-plant-water relations and water footprint of cotton and sorghum under minimum tillage and cover crop rotations. Annual meeting, ASA-CSSA-SSSA. Nov 9-13, 2024, San Antonio, TX.
13. Joshi, P., **M. Dhakar**, D. Spencer, Z. Reynolds. 2025. Strategic fertility management reduces greenhouse gas emissions in corn. Southern ASA conference, Feb 1-4, 2025, Irving, TX.
14. Oloyede, O.E., P. Joshi, D. Spencer, J. Krutz, **M. Dhakar**. 2025. Effect of alternate wetting and drying with automated tailwater reuse on greenhouse gas emissions in rice production. Graduate Research Symposium, Mississippi State University, Feb 15, Starkville, MS.
15. Joshi, P., **M. Dhakar**, D. Spencer, Z. Reynolds. 2025. Strategic fertility management reduces greenhouse gas emissions in corn. Graduate Research Symposium, Mississippi State University, Feb 15, Starkville, MS.
16. **Dhakar, M.**, P.P. Joshi, D. Spencer, Z. Reynolds, J. Krutz. 2024. Mitigating greenhouse gas emissions using improved N management strategies in corn. American Geophysical Union annual meeting. 9–13 December 2024, Washington D.C.
17. **Dhakar, M.**, M.A. Locke, K.N. Reddy, M. Moore, W. Steinriede, J. Krutz. 2024. Soil-plant-water relations and water footprint of cotton and sorghum under minimum tillage and cover crop rotations. Annual meeting, ASA-CSSA-SSSA. Nov 9-13, 2024, San Antonio, TX.
18. Lunden, C., L. Archer, **M. Dhakar** et al. 2024. Efficacy and effects of organic groundcover management strategies on weed suppression, soil health, and pest pressure in mid-Atlantic apple orchards. Annual meeting, ASA-CSSA-SSSA. Nov 9-13, 2024, San Antonio, TX.
19. Tulsi Kharel, H.L. Tyler, P. Mubvumba, Y. Huang, A.B. Bhandari, R.S Fletcher, S. Anapalli, D.R. Joshi, A. Mengistu, G. Birru, K. Adhikari, **M. Dhakar**, K.N. Reddy, D.E. Clay. 2024. Mixed species cover crop and nutrient yield estimation using multispectral drone imagery. Annual meeting, ASA-CSSA-SSSA. Nov 9-13, 2024, San Antonio, TX.
20. **Dhakar, M.**, M.A. Locke, K.N. Reddy, M. Moore, W. Steinriede, J. Krutz. 2024. Water footprint of cotton and sorghum production under conservational management practices. Water Resources Conference 2024. Flowood, MS. Oct 9-11, 2024.
21. **Dhakar, M.**, C. Garrity, A. Smith. 2023. Row spacing and plant density affect weed dynamics, yield, and quality of cover crop-based no-till organic soybean. Annual meeting, ASA-CSSA-SSSA. Oct 29–Nov 1, 2023, St. Louis, MO.
22. Ghalegholabbehbahani, A., G. Zinati, **M. Dhakar**, D. Pandey. 2023. Application of ultraviolet light and milstop to restrict powdery mildew infestation in organic vegetable greenhouses. Annual meeting, ASA-CSSA-SSSA. Oct 29–Nov 1, 2023, St. Louis, MO.
23. **Dhakar, M.** 2023. Industrial hemp production in cold climatic zone. Specialty Crop Symposium, Rodale Institute Pocono Organic Center. 23 August 2023, Blakeslee, PA.
24. **Dhakar, M.**, R. Afshar, C. Garrity. 2023. Integrating electrocution as a supplemental weed

control tool in no-till organic soybean. Weed Science Society of America annual meeting 2023, 30 Jan - 2 Feb 2023, Arlington, VA.

25. **Dhakal, M.**, C. Lapham, R. Afshar. 2022. Effect of tillage, cover crop, and N application rates on hemp production. Annual meeting, ASA-CSSA-SSSA. 6–9 November 2022, Baltimore, MD.
26. **Dhakal, M.** M.A. Locke, K. Reddy, and M. Moore. 2021. Tillage mediates effects of winter cover crop on soil water relationships. Annual meeting, ASA-CSSA-SSSA. 7–10 November 2021, Salt Lake City, UT.
27. **Dhakal, M.**, M.A. Locke, K. Reddy, and M. Moore. 2021. Soil water dynamics and use by cover crops as modified by tillage in cotton and sorghum production systems. USDA-ARS Postdoc Network Fall Seminar Series. 7 October 2021.
28. **Dhakal, M.**, M.A. Locke, M. Moore, K. Reddy. 2020. Cover crop and tillage influence on growth, yield, and soil-plant-water relation of cotton and sorghum. Annual meeting, ASA-CSSA-SSSA. 9–12 Nov. 2020 virtual.
29. **Dhakal, M.**, M.A. Locke, K. Reddy, Y. Huang, and M. Moore. 2020. Growth, yield, and soil-plant-water status of cotton and sorghum under different crop and tillage systems. 75th International Annual Conference. Soil Water Conservation Society. Virtual (<https://bit.ly/SWCSWEB>). 27–29 July 2020.
30. **Dhakal, M.**, and C.P. West. Trade-off between forage quality improvement and crop water use for alfalfa-grass system. Annual meeting, ASA-CSSA-SSSA, San Antonio, TX. 9–13 Nov. 2019.
31. **Dhakal, M.**, and C.P. West. Trade-off between native grassland improvement and water consumption. 2019 American Society of Agronomy Southern Regional Branch. 1–5 Feb. 2019.
32. **Dhakal, M.**, C.P. West. 2018. Trade-off between forage improvement and water use for grass-alfalfa systems. Three Minute Thesis Competition. 26 October 2018. Texas Tech University Graduate School, Lubbock, TX.
33. **Dhakal, M.**, and C.P. West. Establishment and persistence of alfalfa no-till interseeded into semiarid native grassland. Annual meeting, ASA-CSSA, Baltimore, MD. 4–7 Nov. 2018.
34. **Dhakal, M.**, C. P. West. 2017. Improvement of native grassland with alfalfa. 13th Annual Advancing Teaching and Learning Conference, Teaching, Learning, and Professional Development Center, Texas Tech University. 03/03/2017.
35. **Dhakal, M.**, and G. Kharel. Weed management on direct-seeded rice. Annual meeting, ASA-CSSA-SSSA, Tampa, Florida. 22-25 Oct. 2017.

E. Abstracts Presented – Poster

1. Joshi, P.P., O. Oloyede, G.D. Spencer, **M. Dhakal**. (2025). Can protected fertility management mitigate greenhouse gas emission and improve corn productivity? CANVAS annual meeting, ASA-CSSA-SSSA. Nov 9-13, 2025, Salt Lake City, UT (*Third place*).
2. Shrestha, A., T.P. Kharel, **M. Dhakal**, K.N. Reddy. 2025. Satellite-based analysis of spatial and temporal trends in soybean yield across the Lower Mississippi River Basin. CANVAS annual meeting, ASA-CSSA-SSSA. Nov 9-13, 2025, Salt Lake City, UT.
3. Mendez, K., Quiñones, C. Larazo, N., Larazo, W., Adviento-Borbe, M.A.A., **Dhakal, M.**, Massey, J.H., Krutz, J. Greenhouse Gas Emissions from Furrow Rice using Tailwater Recovery System. 2025 Rice Technical Working Group Conference, Feb 17-20, 2025, New Orleans, LA.

4. Joshi, P., K. Mendez, A. AdvientoBorbe, D. Spencer, Z. Reynolds, **M. Dhakal**. 2025. Automated furrow irrigated rice production system mitigates greenhouse gas emissions. Graduate Research Symposium, Mississippi State University, Feb 15, Starkville, MS.
5. Oloyede, O.E., P. Joshi, D. Spencer, J. Krutz, **M. Dhakal**. 2025. Developing a model to estimate evapotranspiration and methane flux using machine learning and remote sensing data. Graduate Research Symposium, Mississippi State University, Feb 15, Starkville, MS.
6. Tulsi Kharel, H.L. Tyler, P. Mubvumba, Y. Huang, A.B. Bhandari, R.S Fletcher, S. Anapalli, D.R. Joshi, A. Mengistu, G. Birru, K. Adhikari, **M. Dhakal**, K.N. Reddy, D.E. Clay. 2024. Mixed species cover crop and nutrient yield estimation using multispectral drone imagery. Annual meeting, ASA-CSSA-SSSA. Nov 10-13, 2024, San Antonio, TX.
7. **Dhakal, M.**, M.A. Locke, K. Reddy, M. Moore, W. Steinriede, L.J. Krutz. 2024. Improving soil water storage with no-till cover cropping in the Mississippi River Alluvial Basin. Bouyoucos Summer Conference, SSSA. 10–12 June 2024, San Juan, Puerto Rico.
8. Philips, H.O., A. Benson, **M. Dhakal**, Y. Rui, A. Smith. 2023. Effectiveness of hairy vetch mechanical termination methods in rotational no-till organic corn. Annual meeting, ASA-CSSA-SSSA. Oct 29–Nov 1, 2023, St. Louis, MO.
9. **Dhakal, M.**, C. Lapham, R. Afshar. 2022. Agronomic strategies to integrate hemp in organic cropping systems. The Pennsylvania Hemp Summit, 14-15 November 2022, Harrisburg, PA.
10. Pandey, D., **M. Dhakal**, C. Lapham, R. Afshar, A. Ghalehgolabbehbahani, E. Omondi, T. Valentine, F. Etemdai. 2022. Optimizing Soil Fertilization and Weed Management with the Aim of Improving CBD-Hemp Yield and Quality in Regenerative Organic Agroecosystems. ASA, CSSA, SSSA Int. Annual Meeting, Baltimore, MD.
11. Maples, H., G. Kharel, **M. Dhakal**, A. Singh, E. Escamilla, S.K. Deb. 2021. Effects of long-term perennial and annual pasture systems on soil physical and hydraulic properties in a semiarid environment. Annual meeting, ASA-CSSA-SSSA. 7–10 November 2021, Salt Lake City, UT.
12. **Dhakal, M.**, Y. Huang, M.A. Locke, K. Reddy, M. Moore, and J. Krutz. 2020. Estimation of cotton and sorghum crop density and cover at early vegetative stages using unmanned aerial vehicle imagery. American Geophysical Union annual meeting. 1–17 Dec. 2020 virtual.
13. **Dhakal, M.**, M.A. Locke, K. Reddy, M. Moore, and W. Steinriede. 2020. Tillage and winter cover crop affects soil-plant-water relationships in sorghum and cotton. Annual meeting, ASA-CSSA-SSSA. 9–12 Nov. 2020 virtual.
14. **Dhakal, M.**, C.P. West, and J.O. Sarturi. 2019. Trade-off between forage quality improvement and crop water use for alfalfa-grass system. 10th Texas Tech Annual Biological Sciences Symposium. 26-27 April 2019. Texas Tech University, Lubbock, TX.
15. **Dhakal, M.**, C.P. West, and S.K. Deb. 2018. Can interseeded alfalfa deplete water in semi-arid grasslands? Annual meeting, Am. Soc. Agron. and Crop Sci. Soc. Am., 4 – 7 Nov 2018, Baltimore, MD.
16. **Dhakal, M.**, C.P. West, and S.K. Deb. 2018. Can interseeded alfalfa deplete water in semi-arid grasslands? International Aridlands Conference, 13–14 August 2018, Office of International Affairs, Texas Tech University, Lubbock, TX.
17. **Dhakal, M.**, C.P. West, C. Villalobos. 2018. Improvement of native grassland with drought-tolerant alfalfa. 17th annual graduate school poster competition, Texas Tech University, Lubbock, TX. 10 April 2018.
18. **Dhakal, M.**, and C.P. West. 2017. Improvement of native grassland with drought-tolerant

alfalfa in the Southern High Plains. 16th annual graduate school poster competition, Texas Tech University, Lubbock, TX. 12 March 2017.

F. Other publications/communications (mass media, digital media)

1. **Dhakal, M.**, Locke, M. A., Reddy, K. N., Moore, M. T., Steinriede, R. W., & Krutz, L. J. (2024). No-till improves soil water storage in Mississippi Delta. *CSA News* 69(6): pp20. <https://doi.org/10.1002/csan.21300>
2. West, C.P., K. Radicke, **M. Dhakal**, L. Baxter, P. Brown, P. Green, and R. Kellison. 2020. Alfalfa in low-irrigation and rainfed grass-based pastures in West Texas. Rancher's Thursday Lunchtime Series, August 13, 2020. <https://www.youtube.com/watch?v=iJqa79QQJC4>
3. Video production about an overview of NARDF funded projects in Nepal. <https://www.youtube.com/watch?v=AXhxARYC-Eg>

G. Webinars

1. "Weed management in rotational organic no-tillage systems", 3/08/2023, Rodale Institute. Available at <https://youtu.be/oBkVY3Gyn0A>.
2. "Growing Industrial Hemp as a multipurpose crop in organic cropping systems", 05/10/2023, Rodale Institute. Available at <https://www.youtube.com/watch?v=Czafdhayk20&t=1s>

H. Scientific Blogs

1. Dhakal, M. 2022. How does young soil support plant life? In: Soils matter, get the scoop! Publisher: ASA-CSSA-SSSA, Madison, WI. <https://soilsmatter.wordpress.com/2022/08/15/how-does-young-soil-support-plant-life/>

I. Invited Professional Talks/Roundtable Discussions

1. "Pathways to sustainable soil and water management: A multi-scale approach to understanding soil-plant-atmospheric continuum". Multistate research (NC-1178) project meeting, June 2-3, 2025, Knoxville, TN.
2. "Calculating water footprint of crop production". Long-Term Agroecosystem Research (LTAR) project network meeting series. Via zoom on Jan 15, 2025.
3. "Integrated weed management program at Rodale Institute". Research seminar presentation on 8/25/2022 at Rodale Institute.
4. "Industrial hemp production in cold climatic zone". Specialty Crop Symposium, Rodale Institute – Pocono Organic Research Center, Blakeslee, PA. 23 August 2023.
5. "Weed management in organic crops". Lecture to Rodale Institute's Farmers' Training participants at Rodale Institute, 7/25/2022.
6. "Understanding soil-plant-water relationships for irrigation". Lecture to Rodale Institute's Farmers' Training participants at Rodale Institute, 8/12/2022.
7. "Organic weed management in row-crops and pastures". Lebanon Grazing Conference 2022, Lebanon, Pennsylvania. Lebanon Forage and Grassland Coalition. 20 January 2022.
8. "Soil water dynamics and use by cover crops as mediated by tillage". USDA Postdoc Network Seminar. October 4, 2021.
9. "Improvement of native grassland with alfalfa". Plant and Soil Science department seminar.

Texas Tech University. 11/26/2018.

10. "Trade-off between forage improvement and water use for grass-alfalfa systems". Three Minute Thesis Competition. 26 October 2018. Texas Tech University Graduate School, Lubbock, TX.
11. "Translational research to create local solutions". 2016 Future Leaders Forum. Association for International Agricultural Research and Development. Annual meeting, 7/10/2016, Washington D.C.

Grant Writing Experiences :

Total grant amount secured: US \$2,328,219

Awarded Projects

Project title	Funding Source	Role	Grant Number	Budget (US\$)*	Year
Rapid on-farm assessment of soil nitrate level using colorimetric technique in corn	MS Corn Promotion Board	PI	Pending contract	53,389	2026-2027
Tailwater capture and reuse: A technological intervention enabling sustainable, climate smart rice production	USDA-NIFA-AFRI	Co-PI	Pending contract	550,000	2026-2029
Polysulphate amendments to reduce methane emissions from rice fields	ICL Growing Solutions	PI	Non-restricted	24,995	2025
Identify agronomic interventions need for climate-smart agriculture using remote sensing, cloud computing, machine learning and big data approach	USDA-ARS	PI	NACA-58-6066-4-027	192,992	2024-2029
Living mulch and grazing techniques to improve soil health and weed control for farmers transitioning to organic farming across climatic zones	USDA-NIFA-ORG	Co-PI	USDA-NIFA-ICGP-009747	1,000,000	2024-2027
Weed management strategies for organic high-density apple orchards	Pennsylvania SCBG – PA Dept. Agric.	Co-PI		127,208	2023-2026
Maximizing the effectiveness of cover crop rolling/crimping for improved weed suppression in no-till organic systems	USDA-SARE Northeast	PI	LNE23-473R-AWD-00001024	199,999	2023-2026
Application of ultraviolet light and MilStop to restrict powdery mildew infestation in vegetable greenhouses	USDA-SARE Northeast	Co-PI	860677	28,131	2022-2023
Bioherbicide for weed management in no-till organic crop production system	Harpe Bioherbicide Inc.	PI		150,000	2022-2024
Weed management on direct-seeded rice (MS thesis research)	Int. Maize and Wheat Improv. Center (CIMMYT)	PI		1500	2009-2011

Mentorship

No. of Mentee(s)	Position	Organization	Duration
2	Ph.D. GRAs (Chair)	Mississippi State University	2024-date
1	MS GRA (Member)	Mississippi State University	2025-date
2	Postdoctoral	USDA-ARS	2025-date

	Associate		
3	Undergraduate Intern	Mississippi State University	2024-date
5	Golden Opportunity Scholars	ASA-CSSA-SSSA	2024-2025
1	Postdoctoral mentee	ARS Postdoc Network	2020-2021
5	Student Intern	Rodale Institute, Kutztown, PA	3 months (2022, 2023)

Professional Development Training

1. **HYDRUS 2025** - “Modeling of water flow and contaminant transport in porous media using the HYDRUS software package.”, by the UC Riverside, and PC-Progress, Ltd, Prague, Czech Republic. Sept 2-3, 2025. (Trainer: Dr. Jirka Šimůnek).
2. **DSSAT 2024**-International Training Program on “assessing crop production, water and nutrient management, climate risk and environmental sustainability with simulation models. May 20-25, 2024, by Hoogenboom et al. at the Uni. of Georgia, Griffin, GA.
3. **Applied Machine Learning in Python**, four online modules (**34 h**) from COURSERA (www.coursera.org), Uni. of Michigan. Apr – Dec 2020.
4. **Deep Learning**, five online courses (**79 h**) from COURSERA (www.coursera.org), www.deeplearning.ai (Apr – Dec 2020).
5. **Practical Machine Learning**, 4-week online module (**9 h**) from COURSERA (www.coursera.org), John Hopkins Uni., Apr – Dec 2020.
6. **The Groundwork Program**, 11 to 13 January 2017 (**3 days**), organized by Teaching Learning and Professional Development Center of Texas Tech University Library, TX.
7. **Effective Communication Technique**, August 2017 to May 2018 (**9 days**), by Center for Agricultural Science and Communication (CASC), Texas Tech Uni., Lubbock, TX.
8. Numerous trainings on topics such as **Research Strategies, Poster Presentation Design, Oral Presentation** by Texas Tech Uni., Lubbock, TX from 2015 to 2019.
9. **Advanced Course on Governance and Development**, 20 June to 20 July 2015 (**30 days**), sponsored by Nepal Administrative Staff College, Kathmandu, Nepal.
10. **Advanced Course on ArcGIS and Data Management**, 4–10 July 2014 (**7 days**), sponsored by Govt. of Nepal & Asian Development Bank, Biratnagar, Nepal.
11. **Training of Trainers (ToT)**- Integrated Pest Management training on Rice, 13 May to 13 August 2013 (**90 days**) sponsored by Govt. of Nepal, Kathmandu, Nepal.
12. **Result Based Monitoring Framework (RBMF)**, 16–25 September 2012 (**10 days**), sponsored by Govt. of Nepal and Asian Development Bank, Dhankuta, Nepal.

Professional Affiliations and Leadership

1. Co-Chair, SSSA Soils and Environmental Quality Division (2027-2029)
2. Chair, Scientists and Students of Nepali Origin (SSNO), 2026-2027
3. Executive Member, Association of Nepalese Agricultural Professionals of Americas (NAPA), 2026-2028
4. Member, Student Honor Code Council, Mississippi State University (2024-2027)
5. Member, Faculty Grievance Committee, Mississippi State University (2025-2027)
6. Advisory board member, Nepalese Student’s Association of MSU
7. Panelist, USDA-NIFA in 2022, 2023, 2026
8. Member, Regenerative Organic Rice Task Force, 2023 to 2024
9. Member, International Agronomy Award Committee, Am. Soc. Agron., 2023 to 2024.
10. Weed Science Society of America, member, 2022 to date
11. American Geophysical Union, member, 2020 to date

12. *Agronomy Journal* Assoc. Editor Board, Am. Soc. Agron., 2020 to 2023
13. ASA–CSSA–SSSA, member, 2016 to date
14. Member, Editorial Board Earth and Space Science Open Archive (ESSOAr), 2021 to date
15. USDA-ARS, Postdoc Network. 2020 to 2021
16. Soil and Water Conser. Society, Ankeny, IA. <https://www.swcs.org/>. 2020 to date
17. Association of Nepalese Agricultural Professionals of Americas, Life Member since 2019
18. Gamma-Sigma-Delta, The Honor Society of Agriculture, Texas Tech University, Member, 2017 to 2019
19. Agronomy Society of Nepal, Member since 2012; and Horticulture Society of Nepal, Member since 2015
20. Association of International Agriculture and Rural Development, Member, 2016
21. Government of Nepal, Ministry of Agriculture Development, Officer, 2011 to 2015
22. Member, Nepal Red Cross Society, Rampur Campus Unit, 2005 to 2010
23. Member, Non-Resident Nepalese Association of America, 2020 to date

Major Scientific Conferences

1. CANVAS, ASA-CSSA-SSSA annual meeting, 9–12 Nov 2025, Salt Lake City, UT
2. 2025 SWCS International Annual Conference. Aug 3-6, 2026. Soil and Water Conservation Society. Costa Mesa, California.
3. ASA Southern Society Meeting, 1-4 Feb 2025, Irving, TX
4. American Geophysical Union annual conference, 9-13 Dec 2024, Washington DC
5. CANVAS, ASA-CSSA-SSSA annual meeting, 9–13 Nov 2024, San Antonio, TX
6. Weed Science Society of America annual meeting, 30 Jan – 2 Feb 2023, Arlington, VA
7. Pennsylvania Hemp Summit 2022. 14 Nov 2022, Harrisburg, PA
8. ASA-CSSA-SSSA annual meeting, 6–9 Nov 2022, Baltimore, MD
9. 2022 Sust. Agron. Conf., July 19 & 26, Aug. 2 & 9. Am. Soc. Agron. Virtual Webinar.
10. ASA-CSSA-SSSA annual meeting, 7–10 Nov 2021, Salt Lake City, UT
11. 2021 Sust. Agron. Conf., July 20 & 27, Aug. 3, 10, 17, & 24. Am. Soc. Agron. Virtual Webinar.
12. ASA-CSSA-SSSA annual meeting, 9–12 Nov 2020, virtual.
13. 2020 SWCS International Annual Conference. July 27-29, 2020. Soil and Water Conservation Society, IA, USA.
14. ASA-CSSA-SSSA annual meeting, 9–13 Nov 2019, San Antonio, TX.
15. Association of International Agriculture and Rural Development Future Leaders Lesson's Learned Workshop, 16-17 September 2019. University of Tennessee, Knoxville.
16. 2019 ASA Southern Branch Meeting 2019, 1–5, Feb 2019, Birmingham, AL.
17. ASA-CSSA-SSSA annual meeting, 4–7 Nov 2018, Baltimore, MD.
18. International Aridlands Conference, 13 – 14 August 2018, Office of International Affairs, Texas Tech University, Lubbock, TX.
19. Sustainable Rangeland Symposium, 7–8 June 2018, National Ranching Heritage Center, Lubbock, TX.
20. Ogallala Aquifer Program Workshop, 27–29 March 2018, Lubbock, TX.
21. ASA-CSSA-SSSA annual meeting, 22 – 25 October 2017, Tampa, Florida.
22. 13th Annual Advancing Teaching and Learning Conf., 3 March 2017, Teaching Learning

and Professional Develop. Cent., TTU, Lubbock, TX.

23. 52nd annual meeting of Association for International Agriculture and Rural Develop. and 2016 Future Leaders Forum, 5–9 June 2016, Washington DC.
24. International Conference on *Environment and Agriculture* (INCEA-98), 1–3 November 2012, Ecological Society of Nepal, Kathmandu, Nepal.
25. 21st Symposium on *Securing sustainable livelihood in the Hindu Kush Himalayas: directions for future research, development, and cooperation*, 5–6 December 2009, Int. Cent. for Integrated Mountain Develop. (ICIMOD), Kathmandu, Nepal.
26. Regional Conference on *Climate Change and Food Security*, 15–18 November 2009, Food and Agric. Org. of the United Nation, Kathmandu, Nepal.

Volunteering and Social Services

1. Judge: Student's oral and poster presentations at ASA-CSSA-SSSA annual meetings from 2020-2025, AGU annual conference 2025, MS Water Resources Conference 2024, and graduate and undergraduate research symposiums at Mississippi State University
2. Session moderator: ASA-CSSA-SSSA meetings, MS Water Resources Conference 2024
3. Reviewer: Foundation for Food and Agriculture Research (FFAR), 2022 and 2023
4. Review panelist, USDA-NIFA Alfalfa Seed and Alfalfa Forage System, 2023
5. Review panelist, USDA-NIFA Tribal Colleges Extension Programs, 2022
6. Review panelist for 'Organic Agriculture Centre of Canada', 2022
7. Associate Editor (2020–2023), *Agronomy Journal*
8. Editorial Board Member, Earth and Space Science Open Archive (ESSOAr)
9. Reviewer: Springer *Nature*, *Front. Agron.*, *Agron. J.*, *Crop Sci.*, *J. Environ. Q.*, *Field Crop Res.*, *PLOS One*, and numerous journals
10. Community services: blood donations (Blood Donors Association of America), panel discussion (Nepalese Agricultural Professionals in Americas), fundraising (Nepal Red Cross Society), and donations to several small charities.
11. Participating in cultural and recreational programs through the Non-Resident Nepalese Association of America and local associations.